

VA code	VA title	Guidelines
VAs-01 Infectious and parasitic diseases		
VAs-01.01	Sepsis	1. Fever AND mental confusion for less than 1 week AND one of the following: • Inability to stand up (<i>information needs to come from the open narrative</i>) • Lethargy (<i>information needs to come from the open narrative</i>). NOTE: If there is a more specific cause, like pneumonia, that would be the underlying cause. Sepsis in those cases would be considered the direct cause.
VAs-01.02	Acute respiratory infection, including pneumonia	1. Acute cough (dry or productive) AND fever AND any of the following: • Fast breathing/breathlessness • Chest pain • Cough up blood (i.e., blood in sputum) • Wheezing. AND none of the following: • Swelling of legs • Distension of abdomen.

VAs-01.03 **HIV/AIDS related death**

1. H/O severe weight loss in less than 3 months (*information on duration needs to come from the open narrative*) **AND** HIV +ve serology.

OR

2. H/O severe weight loss in less than 3 months (*information on duration needs to come from the open narrative*) **AND** any two of the following:
 - History of prolonged unexplained fever
 - Persistent diarrhoea
 - Persistent cough for more than 1 month (intermittent or continuous)
 - Mouth sores/white patches in mouth
 - Skin rash for prolonged periods of time
 - Generalised swelling of nodes (i.e., lumps) in armpits, neck, groin
 - History of spouse/partner with similar illness/death of spouse partner from similar illness (*information needs to come from the open narrative*).

VAs-01.04 **Diarrheal diseases**

1. Three or more loose/liquid/watery stools per day over < 2 weeks **AND** any two of the following:
 - Eyes sunken/depressed fontanelle
 - Low/nil urine output
 - Vomiting
 - Intravenous fluid intake/ORS
 - Mental confusion **OR** dizziness (*information needs to come from the open narrative*)
 - Unconsciousness/lethargy (*information needs to come from the open narrative*)
 - Decreased fluid intake (*information needs to come from the open narrative*).

OR

2. Three or more loose/liquid/watery stools per day over < 2 weeks **AND** blood in stool.
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VAs-01.05 **Malaria**

1. Acute onset of severe fever (may be intermittent) **AND** NO stiff neck **AND** NO bulging of fontanelle (for children under 12 months old) **AND** any three of the following:

- Convulsions
- Unconscious
- Breathlessness
- Pallor (pale palms, eyes, nail beds)
- Jaundice
- Severe headache
- Chills/rigor (*information needs to come from the open narrative*).

OR

2. Acute onset of severe fever (may be intermittent) **AND** recent test positive for malaria **AND** any two of the following:

- Convulsions
- Unconscious
- Breathlessness
- Pallor (pale palms, eyes, nail beds)
- Jaundice
- Severe headache
- Chills/rigor (*information needs to come from the open narrative*).

NOTE: Presence of symptoms of ARI, diarrhoea or burning during micturition could be suggestive of immediate or contributory causes.

VAs-01.06 **Measles**

1. Age $\geq$ 4 months **AND** rash all over body $\geq$ 3 days **AND** fever $\geq$ 3 days **AND** any of the following:

- Rash started on the face (measles rash)
 - Cough
 - Red or watery eyes (*information needs to come from the open narrative*)
 - Running nose (*information needs to come from the open narrative*).
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VAAs-01.07	Meningitis and encephalitis	1. (Continuous fever OR hypothermia in children < 3 months old (<i>information needs to come from the open narrative</i>)) AND any two of the following: • Neck stiffness • Convulsions • Bulging fontanelle • Severe headache • Prolonged loss of consciousness • Vomiting/ Poor feeding (children) • Mental confusion • Irritability or lethargy (<i>information needs to come from the open narrative</i>) • Ear discharge (<i>information needs to come from the open narrative</i>).
VAAs-01.08	Tetanus ³¹	1. (Stiffness of the whole body AND/OR unable to open the mouth) AND any of the following: • Arching of the back/whole body • History of injury (cut or pierced) • Difficulty breathing.
VAAs-01.09	Pulmonary tuberculosis	1. Chronic cough of ≥ 2 weeks AND fever AND any two of the following: • Cough up blood (i.e., blood in sputum) • Chest pain • Breathlessness • Diagnosis and/or treatment history of TB • Known HIV positive serology (especially if not on ART) • Loss of appetite (<i>information needs to come from the open narrative</i>) • Chronic weight loss (<i>information on duration needs to come from the open narrative</i>).

³¹ Excludes: Neonatal tetanus VAs-10.05

VAs-01.10	Pertussis	- Severe cough AND any of the following: “whooping” sound when coughing - Stridor - Vomiting after coughing (<i>information needs to come from the open narrative</i>) - Breathlessness in the neonate (<i>information needs to come from the open narrative</i>).
VAs-01.11	Haemorrhagic fever³²	Fever > 3 days AND bleeding from nose, mouth or anus AND blood test for dengue fever negative (<i>information needs to come from the open narrative</i>).
VAs-01.12	Dengue fever	- Sudden onset of severe fever AND any 2 of the following: - Vomiting - Bleeding from nose, mouth, or anus - Severe pain in muscles, bones, forehead and behind eyeballs (<i>information needs to come from the open narrative</i>). OR - Diagnosis of dengue fever by a health professional.
VAs-01.13	COVID-19	- Severe difficulty breathing or breathlessness (i.e., shortness of breath) until death AND positive test for COVID-19.³³ OR - Severe difficulty breathing or breathlessness until death AND not vaccinated against COVID-19 AND any three of the following: - Fever/chills - Loss or change of smell/taste - Extreme fatigue - Cough - Known exposure to COVID-19 - Diarrhoea/vomiting - Sore throat (<i>information needs to come from the open narrative</i>) - Muscle or body aches (<i>information needs to come from the open narrative</i>).

NOTE: The open narrative is very important to elicit the necessary details to make a correct diagnosis for COVID-19.

VAs-01.99	Unspecified infectious disease	1. Fever with no other specific signs or symptoms.
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Non-communicable diseases

NOTE: This group covers all non-communicable conditions. Any infection of the systems that are listed in this section should be assigned to the suitable infectious disease category. Any maternal and perinatal condition should be assigned to the maternal and perinatal causes below.

VAs-98	Other and unspecified non-communicable disease	No specific guidelines. NOTE: This group covers all non-communicable conditions that could not be assigned to another category in this section. There is a separate category for cases where the cause of death is unknown.
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VAs-02 Neoplasms

VAs-02.01	Oral neoplasms	1. (Lump or sores OR mass OR swelling on tongue/ cheek/ mouth cavity/ gum/ palate, usually progressive (<i>information on swelling needs to come from the open narrative</i>)) AND any of the following: • H/O chewing tobacco • Weight loss • Non-healing sore or ulcer (<i>information needs to come from the narrative</i>) • Restriction/difficulty in opening mouth (<i>information needs to come from the open narrative</i>) • Bleeding on touch from the lump/mass/swelling (<i>information needs to come from the open narrative</i>). OR 2. Diagnosed as oral cancer.
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³² Excludes: Dengue VAs-01.12.

³³ Only test for COVID-19 conducted in health professionals are acceptable.

VAs-02.02 **Digestive neoplasms**

Oesophageal cancer

1. Progressive difficulty in taking foods **AND** weight loss over several months (*information on duration needs to come from the open narrative*).

OR

2. Diagnosed as oesophageal cancer.

Stomach cancer

1. Vomiting **AND** mass in upper abdomen (*information on location needs to come from the open narrative*) **AND** any of the following:
 - Blood in vomit
 - Weight loss
 - Difficulty in swallowing
 - Black stools (*information needs to come from the open narrative*)
 - Black vomit.

OR

2. Diagnosed as stomach cancer.

Colorectal cancer

1. Bleeding from anal opening **AND** any of the following:
 - Chronic H/O constipation alternating with loose stools **OR** constipation alone (*information needs to come from the open narrative*)
 - Weight loss
 - Painful abdominal distension
 - Lump in lower part of abdomen (*information on location needs to come from the open narrative*).

OR

2. Diagnosed as colorectal cancer.

Liver cancer

1. Mass in the abdomen in the right upper quadrant (*information on location needs to come from the open narrative*) **AND** any two of the following:

- Protrusion of abdomen developed slowly over months
- Weight loss
- Jaundice.

OR

2. Diagnosed as liver cancer.

VAs-02.03 **Respiratory neoplasms**

Throat cancer

1. (Hoarseness of voice slowly developed over months *(information needs to come from the open narrative)* **OR** swelling inside the throat *(information needs to come from the open narrative)*) **AND** weight loss over several months *(information on duration needs to come from the open narrative)*.

OR

2. Diagnosed as throat cancer.

Lung cancer

1. Chronic cough **AND** coughing blood **AND** NO diagnosis of Tuberculosis during final illness **AND** any of the following:
 - H/O smoking daily over a long period
 - Noticeable loss of weight
 - Recurrent H/O Pneumonia *(information needs to come from the open narrative)*.

OR

2. Diagnosed as lung cancer.

VAs-02.04 **Breast neoplasms**

1. Painless lump in one or both breasts *(information on pain needs to come from the open narrative)* **AND** any of the following:
 - Skin ulceration over breast
 - Enlarged glands (i.e., lumps) in the neck/axilla
 - Discharge from nipple *(information needs to come from the open narrative)*.

OR

2. Diagnosed as breast cancer.
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VAs-02.05	Female reproductive neoplasms	(Vaginal bleeding after menopause OR pre-menopausal foul smelling vaginal discharge with blood (<i>information needs to come from the open narrative</i>)) AND any of the following: - Weight loss - HIV positive/AIDS diagnosis - Bleeding after sex (<i>information needs to come from the open narrative</i>) - HPV positive (<i>information needs to come from the open narrative</i>). OR - Diagnosed as Carcinoma cervix, uterus, ovarian, vulva.
VAs-02.06	Male reproductive neoplasms	(Palpable lesion/Ulcer in the penis of prolonged duration (<i>information needs to come from the open narrative</i>) OR hard swelling/lump on the testicle(s) (<i>information needs to come from the open narrative</i>) OR difficulty in passing urine) AND chronic loss of weight (<i>information on duration needs to come from the open narrative</i>). OR - Diagnosed as prostate, penile, or testicular carcinoma.
VAs-02.99	Other and unspecified neoplasms	No specific guidelines.

VAs-03 Nutritional and endocrine disorders

VAs-03.01	Severe anaemia	- Pallor (palms, eyes, oral mucous membranes) AND NO diagnosis of any cancer/malaria/sickle cell disease/TB/HIV/AIDS/cardiac failure AND any three of the following: - Weight loss - Fatigue/weakness - Breathlessness on exertion - Swollen legs and feet - H/O bleeding anywhere - Dizziness (<i>information needs to come from the narrative</i>).
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OR

2. Diagnosed as Anaemia.

VAs-03.02 **Severe malnutrition**

1. (Not growing properly **OR** losing weight and becoming very thin over months (*information on duration needs to come from the open narrative*)) **AND** any 2 of the following:
 - Recurrent febrile illness
 - Reddish brown discoloration of hair
 - Flaking of skin
 - Pallor
 - Protruding abdomen
 - Swelling of feet
 - Night-blindness (*information needs to come from the open narrative*).

VAs-03.03 **Diabetes mellitus**

1. H/O diabetes **AND** any of the following:
 - Ulcer(s) in the foot over long period (*information on duration needs to come from the open narrative*)
 - Frequent urination (*information needs to come from the open narrative*)
 - Increased thirst/increased hunger (*information needs to come from the open narrative*)
 - Recurrent infection (particularly respiratory) (*information needs to come from the open narrative*).
 - Tingling or numbness of toes (*information needs to come from the open narrative*)
 - Fruity-scented breath (*information needs to come from the open narrative*).

OR

2. H/O diabetes **AND** no other cause of death identified.
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VAs-04 Diseases of the circulatory system

VAs-04.01 Acute cardiac disease³⁴

1. Severe chest pain lasting for more than ½ hour but less than 24 hours **AND** any 2 of the following:
 - Breathlessness (i.e., shortness of breath)
 - Nausea/vomiting
 - Sudden death
 - Pain radiating to left arm back, jaw or shoulders (*information needs to come from the open narrative*)
 - Pain between the shoulder blades (*information needs to come from the open narrative*)
 - Sweating (*information needs to come from the open narrative*).

OR

2. Sudden death with acute breathlessness **AND** (previous H/O heart disease **OR** hypertension).

NOTE: See Section 4.1.1.1 for definition of sudden death.

OR

3. Diagnosed heart attack/Myocardial infarction.

VAs-04.02 Stroke

1. Sudden onset of paralysis of one side of the body in the month preceding death (*information on onset and timing needs to come from the open narrative*) **AND** any of the following:
 - Unconsciousness
 - Headache
 - Loss of vision (*information needs to come from the open narrative*)
 - Urinary incontinence (*information needs to come from the open narrative*)
 - Loss of sensations on one side of the body (*information needs to come from the open narrative*)

³⁴ Includes: Ischaemic heart disease; Pulmonary embolism; Sudden cardiac death; Cardiac arrest, unspecified; Left ventricular failure; and Hypertensive heart disease with heart failure.

- Altered speech (*information needs to come from the open narrative*)
- Face pulled skew (*information needs to come from open narrative*).

NOTE: In long standing cases, commonly leading to development of bed sores with septicaemia, or features of pneumonia as the immediate cause of death.

VAs-04.03 **Sickle cell with crisis**

1. Pallor **AND** fast breathing **AND** H/O joint and bone pain (*information needs to come from the narrative*) **AND** any of the following:
 - Fever
 - Jaundice
 - H/O sickle cell
 - H/O blood transfusion
 - Fainting (collapse) (*information needs to come from the open narrative*)
 - Dizziness (*information needs to come from the open narrative*).

No specific guidelines for this category

VAs-04.99 **Other and unspecified cardiac disease**

VAs-05 Respiratory disorders

VAs-05.01 **Chronic obstructive pulmonary disease (COPD)**

1. Recurrent episodes of productive cough >2yrs **AND** breathlessness, initially episodic (more in winter) later progressive (*information on mode of onset needs to come from the open narrative*) **AND** no H/O TB **AND** any of the following:
 - Ankle swelling late in disease
 - Death is often preceded by signs of terminal pneumonia (*information needs to come from the open narrative*)
 - Wheezing.

NOTE: respondents may suggest that a patient has asthma when referring to COPD, but narrative may indicate adult onset which would be more in keeping with COPD.

VAs-05.02	Asthma	1. Severe acute breathlessness AND H/O asthma AND any of the following: • Usually in children and young adults • Not responding to usual treatment for breathing problems (<i>information needs to come from the open narrative</i>).
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VAs-06 Gastrointestinal disorders

VAs-06.01	Acute abdomen	1. Severe acute abdominal pain AND any of the following: • Vomiting • Fever • Abdominal distension • Jaundice • Constipation (<i>information needs to come from the open narrative</i>) • Rigidity of abdomen (<i>information needs to come from the open narrative</i>) • Back pain that radiates to the buttocks, groin, or legs (<i>information needs to come from the open narrative</i>) • Pallor/coldness in the legs (<i>information needs to come from the open narrative</i>).
VAs-06.02	Liver cirrhosis³⁵	1. Abdominal distension (i.e., fluid in abdomen; ascites) AND jaundice AND swelling of lower limbs AND with or without any of the following: • Vomiting of blood • Passing of blood in stool • H/o chronic alcoholism • Drowsiness/coma (<i>information needs to come from the open narrative</i>).

³⁵ Includes Alcoholic fibrosis/ cirrhosis; Toxic liver cirrhosis; Fibrosis and cirrhosis of liver, excluding alcoholic and toxic, but including 'unspecified liver cirrhosis'

VAs-07 Renal disorders

VAs-07.01 Renal failure

1. H/O renal dialysis/renal failure from health records **AND** no other specific cause of death.

OR

2. (Passing small amount of urine over a long period *(information needs to come from the open narrative)*) **OR** acute onset of decreasing urinary output for more than 1 day *(information needs to come from the open narrative)*) **AND** any two of the following:
 - H/O hypertension
 - H/O diabetes
 - Swelling of the eyelids/face
 - Swelling of the whole body
 - Lower abdominal pain
 - Confusion/drowsiness *(information needs to come from the open narrative)*
 - Hiccups *(information needs to come from the open narrative)*.

VAs-08 Mental and nervous system disorders

VAs-08.01 Epilepsy

1. (H/O epilepsy **OR** H/O generalized convulsions over a long period *(information on duration needs to come from the open narrative)*) **AND** (general convulsions during final illness **AND** loss of consciousness after convulsions) **AND**:
 - No fever
 - No neck stiffness
 - No H/O injury to head up to one week before death *(information needs to come from the open narrative)*.

OR

2. H/O epilepsy **AND** no other specific cause of death.

VAs-09 Pregnancy-, childbirth and puerperium-related disorders

VAs-09.01 **Ectopic pregnancy**

1. Female aged 12-49 years old **AND** sudden severe abdominal pain **AND** (vaginal bleeding **OR** fainted).

NOTE: H/O pregnancy <2 months is supporting evidence but not essential.

VAs-09.02 **Abortion-related death**

1. Female aged 12-49 years old **AND** H/O abortion (i.e., termination before 28 weeks of pregnancy) **AND** died within 42 days of having an abortion **AND** any of the following:
 - Lower abdominal pain and abnormal vaginal discharge
 - Excessive vaginal bleeding
 - Fever until death.

VAs-09.03 **Pregnancy-induced hypertension**

1. (Pregnancy >20 weeks of gestation **OR** up to 6 weeks postpartum) **AND** (diagnosis of high blood pressure by a health professional at >20 weeks gestation **OR** up to 6 weeks postpartum (*information on timing needs to come from the open narrative*)).

OR

2. (Pregnancy >20 weeks gestation **OR** up to 6 weeks postpartum) **AND** any of the following:
 - Convulsions for first time in pregnancy **OR** up to 6 weeks postpartum
 - Swelling of face
 - Blurred vision
 - Severe headache
 - Upper abdominal pain.

VAs-09.04 **Obstetric haemorrhage**

Antepartum haemorrhage

1. Abdominal/back pain after 28 weeks of gestation but before birth of baby (*information needs to come from the open narrative*) **AND** (excessive vaginal bleeding **OR** fainting attack (*information needs to come from the open narrative*)).

OR

2. H/O painless vaginal bleeding during pregnancy (*information on pain needs to come from the open narrative*) **AND** excessive bleeding during labour and delivery.

Postpartum haemorrhage

3. Excessive bleeding after delivery of baby **OR** fainting attack **AND** no abdominal pain **AND** any one of the following:
 - Placenta was partially or fully retained
 - Abdominal pain had stopped after delivering the baby (*information on timing needs to come from the open narrative*).

VAs-09.05	Obstructed labour	1. Abnormal presentation (breech, shoulder, hand, or transverse) AND baby not delivered. OR 2. (Difficulty in delivering baby OR forceps/vacuum/C-section delivery) AND prolonged labour >24 hours.
VAs-09.06	Pregnancy-related sepsis	1. H/O delivery/abortion within 42 days AND severe fever persisting till death AND any of the following: • Foul smelling vaginal discharge • Lower abdominal pain/distention.
VAs-09.07	Anaemia of pregnancy	1. Death during pregnancy, labour, delivery or occurring soon after birth of child AND no other specific cause of death was identified AND any of the following: • (Paleness of tongue, lips or palms OR fatigue) AND breathlessness prior to death • H/O need/given blood transfusion • H/O diagnosis of severe anaemia (HB < 5g/dl) (<i>information needs to come from the open narrative or medical records</i>) NOTE: Other symptoms that can be available from the open narrative include palpitations/tachycardia, irritability, dizziness, and restless legs.
VAs-09.08	Ruptured uterus	1. H/O prolonged labour AND sudden severe abdominal pain AND any of the following: • Vaginal bleeding • H/O c-section.

VAs-10 Neonatal causes of death

VAs-10.01	Prematurity or low birth weight	1. Born alive AND pregnancy duration less than 8 months (i.e., for 7 months and below) AND no other obvious cause of death. OR 2. If pregnancy duration not known: Smaller than average size baby (if weighted, birth weight below 2.5 kilograms) AND no other obvious cause of death.
VAs-10.02	Birth asphyxia³⁶	1. (Delayed or no breathing at birth OR delayed or no crying at birth) AND any sign of life present at birth (i.e., exclude stillbirths) AND any of the following: • Died within 24 hours of birth • Convulsions in first 24 hours • Prolonged or difficult labour • Poor suckling • Respiratory distress after birth in a term infant.
VAs-10.03	Neonatal pneumonia³⁷	1. Breathed immediately at birth, even if a little AND (fever OR cold to touch) AND any of the following: • Fast breathing • Nasal flaring/grunting • Chest in drawing.
VAs-10.04	Neonatal sepsis	1. (Fever OR cold to touch) AND no other obvious causes of death (e.g., acute respiratory infection, including pneumonia, meningitis, diarrhoea) AND any of the following: • Purulent cord • Skin sores/ulcers • Poor suckling • Lethargic • Unconscious • Convulsions.

³⁶ Includes: Hypoxia and respiratory distress

³⁷ Differs from “pneumonia”, as this refers to a death within 28 days of life.

VAS-10.05	Neonatal tetanus	1. Baby able to suckle and cry after birth AND stopped suckling or crying after 2 days AND baby's body became rigid (stiff with the back arched backwards) AND any of the following: • Convulsions • Umbilical cord inflammation (pus draining from the cord or redness extending to the skin) • Mother not vaccinated against tetanus in pregnancy.
VAS-10.06	Congenital malformation	1. Any of the following: • Abnormally small head • Abnormally large head • Swelling or defect of the back • Open palate (cleft) (<i>information needs to come from the open narrative</i>) • No anal opening (i.e., not being able to defecate) (<i>information needs to come from the open narrative</i>) • Defect of the abdominal wall and visible internal organs (<i>information needs to come from the open narrative</i>).
VAS-10.99	Other and unspecified perinatal cause of death	Death within 7 days of birth; No specific guidelines for this category.

VAS-11 Stillbirths

VAS-11.01	Fresh stillbirth	1. Baby did not cry, move, or breathe at birth AND (stopped moving during labour OR no maceration).
VAS-11.02	Macerated stillbirth	1. Baby did not cry, move, or breathe at birth AND (stopped moving before the onset of labour OR baby's body was soft, discoloured and the skin peeling away).

VAS-12 External causes of death

NOTE: The list of questions contains sub questions that allow for more specificity for accidents. Diagnosis is evident from related information from questions and open narrative. Care should be taken to certify external cause as well as body part involved in injury, as applicable.

VAS-12.01	Road traffic accident	
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VAs-12.02	Other transport accident
VAs-12.03	Accidental fall
VAs-12.04	Accidental drowning and submersion
VAs-12.05	Accidental exposure to smoke, fire and flames
VAs-12.06	Contact with venomous animals and plants
VAs-12.07	Accidental poisoning and exposure to noxious substance
VAs-12.08	Intentional self-harm
VAs-12.09	Assault
VAs-12.10	Exposure to force of nature
VAs-12.99	Other and unspecified external cause of death
VAs-99	Cause of death unknown